



Spring 2026

Robots Motion Planning and Control Course 362-2-5481

Course objectives

To learn the analytical and numerical tools and algorithms used to plan and control the motion of robots: both manipulators and mobile robots. To learn various algorithms and approaches for motion planning and analyze their performance. To learn stability analysis method of non-linear dynamical systems, and how to use them for motion control of robots.

Lecturer

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Office: building 55, room 341, or Lab. 4 building 57.

Office hours: as per your request in email or WhatsApp.

Grade

The grade will be given based on a mini-project that will be done at home. The mini-project will have two parts, first to develop a dynamical model of a robot and to implement a motion controller in simulation. The second part is to choose a recent research paper in the area of robot motion planning or control. Then implement the method described in the paper, present it in class, and submit a report.

The mini-project will be done in groups of 1,2, or 3 students.

Teaching program

Week	Topic
1	Introduction to mobile robots - mobile robots structure and SLAM
2	Kinematics of Robots, configuration space, BUG algorithm
3	Road maps, graphs representation, and search algorithms
4	Cells decomposition method, visibility graph, Voronoi diagram, Potential fields
5	Probabilistic road maps, RRT algorithm.
6	Dynamic regrasping, Multi robot search algorithms.
7	Robots Kinematics
8	Robots Jacobian, Virtual work principal, Forces
9	Lagrange's Equations
10	Robot dynamics, Pfaffian's constraints
11	Equilibrium and Lyapunov Stability, Feedback linearization, PD controller
12	Sontag's Formula, Potential functions for motion control
13	Students presentations

Note: the teaching program is only a guideline and there is no commitment to the materials or the timetable.

References

1. Richard M. Murray, Zexiang Li, S. Shankar Sastry, "Mathematical Introduction to Robotic Manipulation", 1994



2. Spong Mark, Hutchinson Seth, Vidyasagar M., "Robot Modeling and Control", John-Wily, 2006.
3. Howie Choset, Lynch, Hutchinson, Kantor, Bugard, Kavarki, Thrun, "Principles of Robot Motion", MIT press, 2005.
4. Spong and Vidyasagar, "Robot Dynamics and Control", John-Wily, 1989.
5. E. Rimon, and J. Burdick, "The Mechanics of Robots Grasping", Cambridge press, 2020.
6. Hassan K. Khalil, "NonLinear Systems", 3rd edition, Prentice-Hall, 2000.
7. STEVEN H. STROGATZ, NONLINEAR DYNAMICS AND CHAOS With Applications to Physics, Biology, Chemistry and Engineering, by Perseus Books Publishing, L.L.C. 1994.
8. Jean-Claude Latombe, "Robot Motion Planning" (The Springer International Series in Engineering and Computer Science) (Hardcover)
9. Steven M. LaValle, "Planning Algorithms".
10. B. Siciliano, O. Khatib, "Springer Handbook of Robotics"
11. "ד"ר ניר שוולב, "מבוא לרובוטיקה"